



**Term Evaluation Theory (Even) Semester Regular Examination February 2026**

Roll no.....

Name of the Course: BTech

Semester: II

Name of the Paper: Basic Electronics Engineering

Paper Code: TEC 201

Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

Q1.

(10 Marks) CO 1

- a. Minimize the following by K map and realize the minimized function by NAND and NOR gate only

$$F(A, B, C, D) = \pi M(0, 1, 3, 5, 7, 8, 9, 11, 13, 14, 15)$$

OR

- b. Simplify the following function with the help of K Map:

$$F(A, B, C, D) = A + BC' + ABD' + ABCD$$

Q2.

(10 Marks) CO 1

- a. Explain min-terms and max-terms. State and prove De-Morgan's law.

OR

- b. Simplify the following functions:

- (i)  $\{(AB' + ABC)' + A(B + AB')\}'$
- (ii)  $A'B'C + (A + B + C)' + A'B'C'D$

Q3.

(10 Marks) CO 1

- a. Perform the number conversion given below:

- (i)  $(111011101)_2 = (?)_{10} = (?)_{16}$
- (ii)  $(88BAE)_{16} = (?)_8 = (?)_{10}$

OR

- b. Simplify following expression and realize the minimized function by NOR gate only

$$F(A, B, C) = A'B'C + A'BC' + A'B + B'C'$$

Q4.

(10 Marks) CO 1

- a. Perform the following number conversion

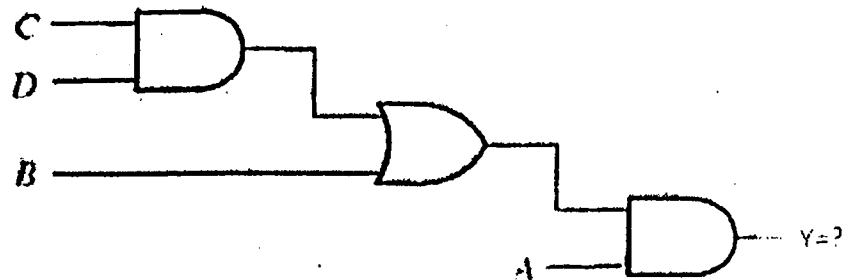
- (i)  $(A9BC2.0C)_{16} = (?)_{10} = (?)_6$
- (ii)  $(432)_x = (225)_7$ , find the base "x".

OR



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- b. Define the output 'Y' of the logic circuit given below and write the truth table for the same.



Q5.

(10 Marks) CO I

- a. Minimize the following function with the help of K map and realize the minimized function by NAND gate only as well as NOR gate only

$$F(A, B, C, D) = \sum m(1, 3, 4, 6, 8, 9, 11, 13, 15) + \sum d(0, 2, 14)$$

OR

- b. Define the output 'Q' of the logic circuit given below and write the truth table for the same. Also, find the min-terms and max-terms.

